NORTH SOUTH UNIVERSITY
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Fall 2026

EEE/ETE 111L.19 Lab | Analog Electronics Lab I

Course Faculty: Dr. Hafiz Imtiaz [hai]

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Office Room: TBA
Office Hours: TBA

[* Email or call to make an appointment beforehand *]

Lab Section: 19
Lab Time: A 11:20 AM – 02:30 PM
Room no: SAC 506

Tentative Marks Distribution

Attendance	10%
Class Performance	5%
Lab Report & Simulation (Individual)	20%
MID Exam	20%
Set-up Test and Simulation	15%
Viva	10%
Final Exam	20%

Tentative Lab Session Schedule

Classes	Lab No	Title of Activities	Date
Class 1	Lab 01	I-V Characteristics of Diode	03/10/2026
Class 2	Lab 02	Half-Wave and Full-Wave Rectifier Circuits	10/10/2026
Class 3	Lab 03	Clipper Circuits and Clamper Circuits	24/10/2026
Class 4	Lab 04	Zener Diode Characteristics and Applications	31/10/2026
Class 5	Lab 05	**Mid Exam** & (The BJT Biasing Circuits)	**14/11/2026**
Class 6	Lab 06	Study of Switching Characteristics of MOSFET	21/11/2026
Class 7	Lab 07	MOSFET Amplifier	28/11/2026
Class 8		**Set-up Test and Simulation**	05/11/2026
Class 9		**Lab Final Exam and Viva**	**12/12/2026**
Class 10			**19/12/2026**

Rules

- Arrive in class on time. Late arrival in the classroom gives a negative impression.
- Don't leave before completing the lab work assigned by the lab instructor. If you need to leave early, ask for permission from the lab instructor.
- Don’t leave the lab room while class is running. If you need to use the washroom, ask the lab instructor for permission. Going to other places, i.e., cafeterias, is not recommended.
- You can talk with the members of your own group and the lab instructor about things related to lab work, but please try not to gossip. Keep your voice low so that it doesn’t disturb others.
- Do not talk on your mobile phones. If you need to make an emergency call, ask for the permission of the instructor and go outside to make or receive the call.
- Copying of lab reports is not accepted.
- All report submissions are due at the beginning of the class.
- Late Submissions will not be accepted without proper reasoning.

Lab Report Writing Format

- Printed Cover page (Edited from Sample)
- Experiment Title
- Objectives
- Theory
- Apparatus List (Tabular Format)
- Circuit Diagrams
- Data Tables and Calculations
- Report (Questions and Answers)
- Result Analysis & Discussion (This is one section, not 2 different sections)
- Attachment (Signed lab sheet) + Simulations

Report Writing Guidelines

After completion of a lab experiment, the lab report is due in the next immediate lab class. Each group has to submit a single lab report for each experiment. One lab report will be written by one person. Late submissions should not be allowed. Below is a detailed description of what each lab report must contain:

1. **Cover Page -** All lab reports should have a cover page, and the same cover page should be used for all the lab reports. A sample of a cover page will be provided to you.
2. **Experiment Title -** Write the title of the experiment clearly.
3. **Objectives** – You should briefly write what the aim of the experiment is. In other words, write what you intend to achieve by experimenting.
4. **Theory (write in your own words)**: Students should briefly explain the theory in their own words. Use proper graphs, diagrams and circuits where necessary.
5. **Apparatus List** – A simple list of all the apparatuses and components you used to do the lab experiment.
6. **Circuit Diagrams** – Draw the circuit diagrams for the experiment; it should be drawn by pencil, but should be clean and legible.
7. **Data Tables and Calculations** – This section of the lab report will contain the data that you have collected practically, and it should be presented in a tabular form (make a fresh, clean table of the data obtained in class). Show all the calculations related to the data table.
8. **Reports (Questions and Answers**) There are a few report questions that you need to answer.
9. **Result Analysis & Discussion** –This is one of the most important parts of the lab report. What you write here proves how attentive and careful you were during the lab class. Copying a single line from another person's discussion or from a previous lab

report will earn you a straight zero.

You must focus on these points:

a. What was the purpose of the experiment?
b. What did you actually do in the experiment?
c. What did you learn from the experiment?
d. Were the data and graph exactly as you expected from the theoretical knowledge, or were the practical results different from theory?
e. If there are differences between theoretical and practical data, what might be the possible reasons?
f. What are some of the problems you faced during the experiment?

10. **Attachment & Simulation** – Finally, you must attach the signed data sheet from the instructor in the lab, and students should include simulation results along with simulated circuit diagrams.

Reminder:

1. Ensure that you attach the **provided cover page** with the lab report. **Edit** the sample **.docx file** uploaded in Canvas with details of your lab.
2. Write the writer's name at the top, **and write "w" beside the writer's name**.
3. Write your name and ID inside the simulation as text.
4. Use graph papers to plot graphs.
5. Attach graph in the segment where it has been asked, not at the end with attachments.
6. Attach the graph paper with the page with multiple stapler pins from both top and side, not from just one side.
7. No copy-paste. All types of copy-paste and sharing reports are strictly prohibited. For any problem, consult with your lab instructor.
